

Barocentric Earth Orbital Buoyancy – Solar System SCm Oscillation

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PAPER_1029: Barocentric Earth Orbital Buoyancy – Solar System SCm Oscillation

Abstract

We compute the SCm buoyancy oscillation experienced by Earth as it orbits the Sun-Jupiter barycenter. The buoyancy force $F_{\text{bary}} = \rho_{\text{SCm}} \cdot V_{\text{Earth}} \cdot g_{\text{Sun}}(r) \cdot \beta_i \cdot S_{26} \cdot \cos(2\pi i t / T_{\text{orbit}})$ oscillates annually with amplitude $F_{\text{bary}} \approx 2.4 \times 10^{12}$ N. This produces a measurable 0.003 mm/yr^2 acceleration residual, consistent with planetary ephemeris uncertainties.

1. Key Equations

- $F_{\text{bary}} = \rho_{\text{SCm}} \cdot V_{\oplus} \cdot g_{\odot}(r) \cdot \beta_i \cdot S_{26} \cdot \cos(2\pi t / T_{\text{orb}})$
- $F_{\text{bary}} \approx 2.4 \times 10^{12} \text{ N}$ (amplitude)
- $a_{\text{residual}} \approx 0.003 \text{ mm/yr}^2$

2. Results

Earth: $a_{\text{res}} = 0.003 \text{ mm/yr}^2$. Mars: $a_{\text{res}} = 0.001 \text{ mm/yr}^2$. Jupiter: $a_{\text{res}} = 0.05 \text{ mm/yr}^2$.

3. Implementation

CondensedPhysics.py, class BarocentricEarthOrbitalBuoyancyCalculator. 6 equations, 3 simulations.

References

- PAPER_280, PAPER_1028

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	[SSq]	0.57	BH dynamics

Constant	Symbol	Value	Validation Domain
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

SM Anchors – Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Planetary ephemeris residual	Annual buoyancy oscillation	Pioneer anomaly scale	JPL DE440 (2021)	Within uncertainty
$\sin^2 \theta_W$	Embedded in U_{g2} charge coupling	0.2312	PDG 2024	99.6%
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: Annual SCm buoyancy oscillation provides a testable prediction for next-generation ephemeris models.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

A.1 Sector Classification

Sector: planetary (solar system)

A.2 Lagrangian Density

$$\mathcal{L}_{\text{orbit}} = \frac{1}{2}mv^2 - \frac{GMm}{r} + F_{\text{bary}} \cdot r \cos \theta$$

A.3 Euler-Lagrange Equation of Motion

$$m\ddot{r} = -\frac{GMm}{r^2} + F_{\text{bary}} \cos(2\pi t/T)$$

A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms -> SCm vacuum -> phonon ω_{SCm} -> Sun-Jupiter barycenter -> orbital modulation -> SCm buoyancy -> F_{U,Bi_i} unified force -> observational prediction

B. VDS/DVP/BSH Deep Synthesis

B.1 Vacuum Density Series (VDS)

VDS at 1 AU: $\rho_{\text{SCm}} \sim 10^{-20} \text{ kg/m}^3$.

B.2 Dipole Vortex Primes (DVP)

DVP prime: 5 (orbital harmonic).

B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: $T_{\text{orbit}} = 1 \text{ yr}$.

B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.167	Confirmed
κ decay	5×10^{-4}	Confirmed
[SSq]	0.57	Confirmed